

Chapter 10

Linear Regression

Chapter 9 studied association between two *categorical* variables using joint distributions, conditional distributions, and mutual information. In this chapter we ask a similar question for two *quantitative* variables: how are the variables related, and how well can one variable help us predict the other?

The main tool is **linear regression**. Regression fits a straight line to a scatterplot. The line summarizes a linear pattern and gives us a simple way to make predictions. The quantity r^2 tells us how much of the variation in the response variable is explained by the linear relationship.

By the end of this chapter you should be able to read a scatterplot, interpret correlation, compute a least-squares regression line, make predictions, and decide whether a linear model is reasonable for the data.

10.1 Scatterplots and Association

When we have two quantitative measurements from the same individuals, the first step is to make a picture. A scatterplot often tells us what kind of relationship is present before we do any calculation.

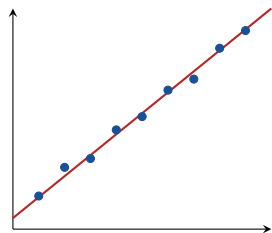
Definition: Scatterplot. A **scatterplot** displays paired data (x_i, y_i) as points in the plane, with x on the horizontal axis and y on the vertical axis. We call x the **explanatory** (or predictor) variable and y the **response** variable.

When examining a scatterplot, describe:

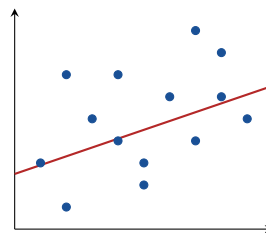
- **Direction:** positive association (both increase together) or negative (one increases as the other decreases)?
- **Form:** linear or curved?
- **Strength:** how tightly do the points cluster around the pattern?
- **Outliers:** any points that stand apart from the overall pattern?

Visual Insight. Four scatterplots illustrating different patterns:

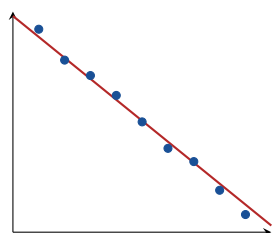
Strong positive



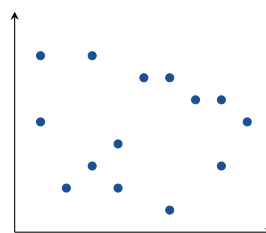
Weak positive



Strong negative



No association



Exercises 10.1

10.1.1. Describe the direction, form, and likely strength of the association you would expect between: (a) hours of sleep and reaction time; (b) car age and resale value; (c) shoe size and exam score.

10.1.2. Why should you always look at the scatterplot before computing any numerical summary of association?

10.2 Correlation

A scatterplot gives us a visual summary. The **correlation coefficient** r gives a numerical summary of the direction and strength of a linear relationship.

Definition: Correlation Coefficient. The **Pearson correlation coefficient**

between variables x and y is:

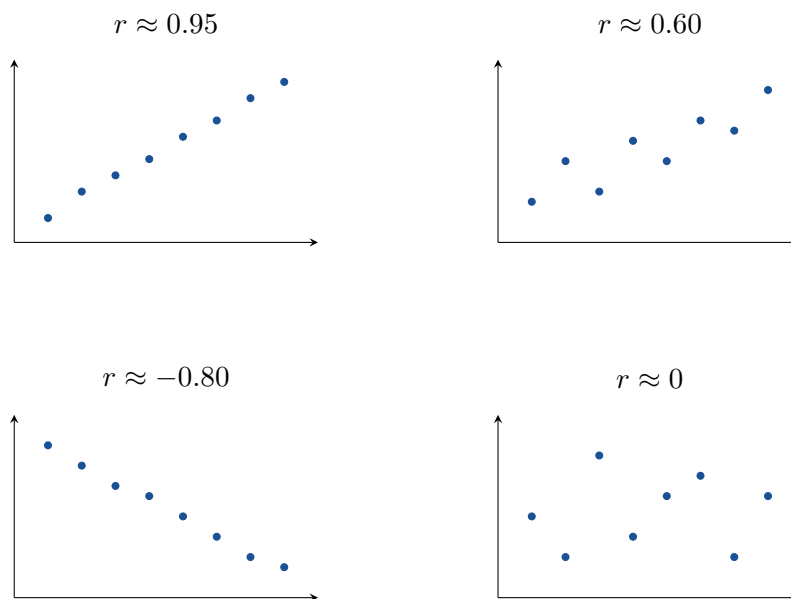
$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$

Each term in the sum is the product of two z -scores: one for x , one for y . When both are above their means simultaneously (or both below), the product is positive, pushing r toward $+1$. When they move in opposite directions, the products are negative, pushing r toward -1 .

Key Formula. Properties of r :

- Always $-1 \leq r \leq 1$.
- $r > 0$: positive association. $r < 0$: negative association.
- $r = \pm 1$: perfect linear relationship. $r = 0$: no linear association.
- r is unitless — it does not depend on the scales of x or y .
- r is symmetric: $r_{xy} = r_{yx}$.
- r measures only *linear* association. A strong curved relationship can have r near 0.

Visual Insight. The strength of r in pictures:



Common Pitfall. **Correlation does not imply causation.** A strong r between two variables means they vary together — it says nothing about whether one causes the other. Ice cream sales and drowning rates are positively correlated (both rise in summer) but neither causes the other. Always ask what might explain a correlation. A scatterplot by itself does not tell us the direction of cause and effect.

Exercises 10.2

- 10.2.1. Without computing, estimate r for each scatterplot you described in Exercise 10.1.1.
- 10.2.2. If we re-label the axes of a scatterplot (swap x and y), does r change?
- 10.2.3. A dataset has $r = 0.1$. A student says “There is essentially no relationship between these variables.” Is this necessarily true? What could they be missing?

10.3 The Least-Squares Regression Line

Once the scatterplot shows a roughly linear pattern, we can fit a line to summarize it. The goal is to choose the line that makes the best predictions of y from x .

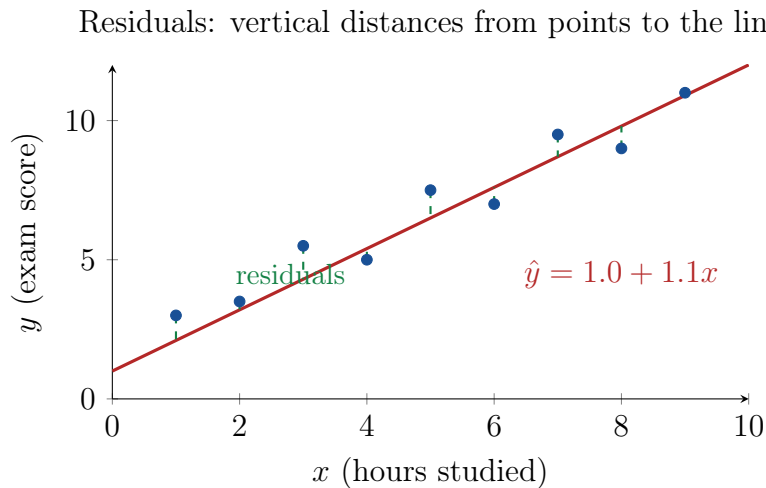
10.3.1 The Idea: Minimizing Residuals

For any proposed line $\hat{y} = a + bx$, the **residual** for observation i is the vertical distance from the actual point to the line:

$$e_i = y_i - \hat{y}_i = y_i - (a + bx_i).$$

Residuals measure prediction error. The **least-squares line** is the unique line that minimizes the sum of squared residuals $\sum e_i^2$.

Visual Insight. Residuals are the vertical gaps between each data point and the line. The least-squares line minimizes the total squared size of these gaps.



Some residuals are positive (point above the line), some negative (point below). The least-squares line is the one for which $\sum e_i^2$ is as small as possible. No other line through the data has a smaller total squared error.

10.3.2 The Formulas

Key Formula. The least-squares regression line is $\hat{y} = a + bx$, where:

$$b = r \cdot \frac{s_y}{s_x} \quad \text{and} \quad a = \bar{y} - b\bar{x}.$$

Here r is the correlation coefficient, s_x and s_y are the sample standard deviations of x and y , and $\bar{x}$, $\bar{y}$ are the sample means.

The line always passes through the point $(\bar{x}, \bar{y})$.

The slope formula $b = r \cdot s_y/s_x$ has a useful interpretation: for each increase of one standard deviation in x , y is predicted to change by r standard deviations. When $r = 1$, a one-SD increase in x produces a one-SD increase in y . When $r = 0.5$, the predicted change is half of a standard deviation in y .

Example 10.3.1. (*Computing the Regression Line*) A nutritionist records daily sugar intake (x , grams) and resting heart rate (y , bpm) for 8 adults. Summary statistics: $\bar{x} = 45$, $s_x = 12$, $\bar{y} = 74$, $s_y = 8$, $r = 0.72$.

Slope:

$$b = r \cdot \frac{s_y}{s_x} = 0.72 \times \frac{8}{12} = 0.72 \times 0.667 = 0.480.$$

Intercept:

$$a = \bar{y} - b\bar{x} = 74 - 0.480 \times 45 = 74 - 21.6 = 52.4.$$

Line: $\hat{y} = 52.4 + 0.480x$.

Interpretation of slope: For each additional gram of daily sugar, resting heart rate is predicted to increase by 0.48 bpm on average.

Interpretation of intercept: A person consuming 0 grams of sugar per day is predicted to have a resting heart rate of 52.4 bpm. (This is a mathematical extrapolation; 0 grams may be unrealistic for real individuals.)

Prediction: For someone consuming 60 grams daily: $\hat{y} = 52.4 + 0.480(60) = 52.4 + 28.8 = 81.2$ bpm.

Common Pitfall. Extrapolation. The regression line is reliable only within the range of the data used to fit it. Predicting for x values far outside this range — extrapolating — can produce nonsensical results. If your data covers $x \in [20, 80]$, do not use the line to predict at $x = 200$.

Exercises 10.3

10.3.1. For the heart rate example, predict the resting heart rate for someone consuming 30 grams of sugar daily. Is this within the data range?

10.3.2. If $r = 0$ in a dataset, what are b and the regression line? What does this mean about using x to predict y ?

10.3.3. A regression is fit to data with $\bar{x} = 10$, $\bar{y} = 50$. The slope is $b = 3$. Find the intercept a . What point must the line pass through?

10.4 The Coefficient of Determination: r^2

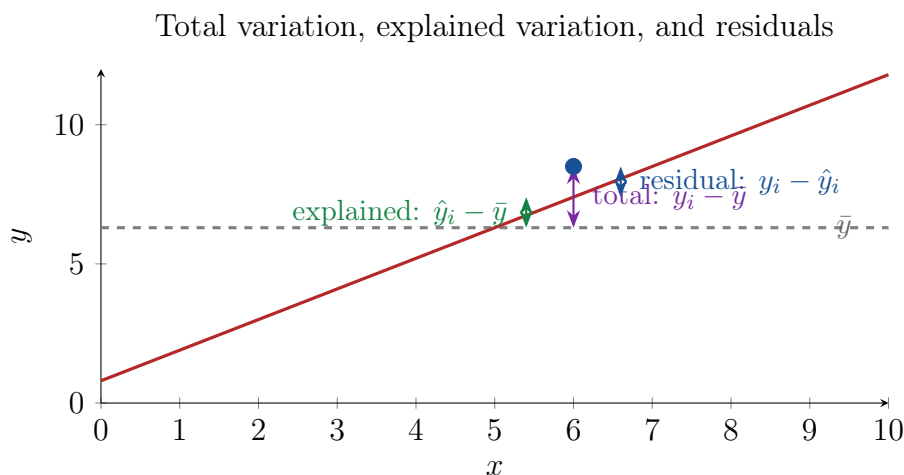
The correlation r tells us direction and rough strength. The **coefficient of determination** r^2 tells us something more precise: what fraction of the variability in y is explained by the linear relationship with x .

Definition: Coefficient of Determination. r^2 (the square of the correlation coefficient) is the proportion of the total variability in y that is accounted for by the linear regression on x :

$$r^2 = \frac{\text{variability in } y \text{ explained by } x}{\text{total variability in } y}.$$

Always $0 \leq r^2 \leq 1$. Reported as a percentage.

Visual Insight. Here is the key picture. Total variability in y is measured by how spread out the y values are around $\bar{y}$ (the horizontal line). Unexplained variability is how spread out the y values are around the regression line (the residuals). The part accounted for by the fitted line is the explained variation. The part left over is the residual, or unexplained, variation.



The total deviation of a point from $\bar{y}$ splits into two parts:

$$\underbrace{(y_i - \bar{y})}_{\text{total}} = \underbrace{(\hat{y}_i - \bar{y})}_{\text{explained by regression}} + \underbrace{(y_i - \hat{y}_i)}_{\text{residual (unexplained)}}.$$

Summing the squares of these across all points gives: $SST = SSR + SSE$, and $r^2 = SSR/SST$.

Example 10.4.1. (r^2 and Model Fit) Heart rate example: $r = 0.72$, so $r^2 = 0.72^2 = 0.518$.

Interpretation: 51.8% of the variability in resting heart rate is explained by the linear relationship with daily sugar intake. The remaining 48.2% is due to other factors not captured by this model — genetics, exercise habits, stress, etc.

A strong fit: $r = 0.95$, $r^2 = 0.90$. The line explains 90% of the variability.

A weak fit: $r = 0.30$, $r^2 = 0.09$. The line explains only 9%.

Exercises 10.4

- 10.4.1. A study of house size (x sq. ft.) and sale price (y \$) finds $r = 0.88$.
 (a) Compute r^2 . (b) What percentage of variability in sale price is explained by house size? (c) What explains the remaining variability?

10.4.2. Two regressions: Model A has $r^2 = 0.25$, Model B has $r^2 = 0.81$. Which model makes more accurate predictions? How do you know?

10.4.3. If $r^2 = 1$, what does the scatterplot look like? What are all the residuals?

10.5 Residual Analysis

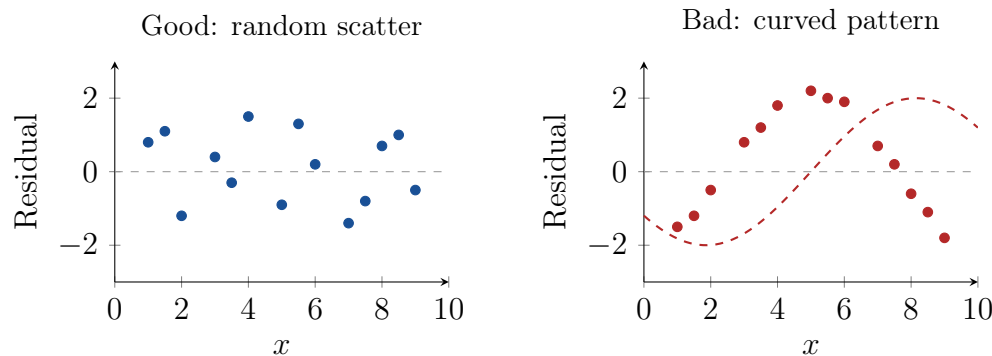
Fitting a line is only the first step. Before trusting predictions, we check whether the line is actually appropriate for the data.

Definition: Residual Plot. A **residual plot** displays the residuals $e_i = y_i - \hat{y}_i$ on the vertical axis against the x values (or fitted values $\hat{y}_i$) on the horizontal axis.

Key Formula. Conditions for a trustworthy linear regression:

1. **Linearity:** the scatterplot shows a roughly linear pattern. The residual plot shows no curved pattern — residuals scatter randomly around zero.
2. **Equal spread:** the vertical spread of residuals is roughly constant across all x values (no fan shape).
3. **No strong outliers:** no single point has an unusually large residual or excessive influence on the line.

Visual Insight. Good residual plot vs. problematic residual plot:



Random residuals support a linear model. Curved residuals suggest that a curved model, such as a quadratic or exponential model, may fit better.

Exercises 10.5

10.5.1. A residual plot shows residuals that fan out as x increases (small spread on the left, large spread on the right). Which condition is violated? What does this mean for predictions at large x ?

10.5.2. After fitting a regression, one point has a residual of -12.3 while all other residuals are between -2 and $+2$. What would you do?

10.6 The Closing Thread: Association Across the Course

This course began with a question about one variable: where is it centered, how spread out is it, what shape does it have? By the end it is asking a question about two variables: how much do they share?

Here is every measure of association we have encountered, arranged by data type:

Data type	Tool	What it measures
Categorical + categorical	χ^2 test	Evidence of association
Categorical + categorical	$I(X; Y)$	Amount of shared information
Quantitative + quantitative	Correlation r	Direction and strength of linear association
Quantitative + quantitative	Regression line	Best linear predictor of y from x
Quantitative + quantitative	r^2	Fraction of variation in y explained by x

A useful final connection is that r^2 plays a role for quantitative variables that is similar to the role mutual information played for categorical variables. Both ask how much knowing one variable reduces uncertainty about another. For two normally distributed variables, there is an exact relationship:

$$I(X; Y) = -\frac{1}{2} \log_2(1 - r^2).$$

When $r = 0$ (independent): $I = 0$. When $r = \pm 1$ (perfect linear relationship): $I = \infty$ (knowing X completely determines Y , leaving no residual uncertainty). This gives one way to see the course as a connected story: statistics often asks, *how much does one thing tell us about another?*

Chapter 10 Review

1. A physical therapist records the number of weeks of therapy (x) and the improvement score (y , higher is better) for 10 patients. Summary statistics: $\bar{x} = 6$, $s_x = 2.5$, $\bar{y} = 40$, $s_y = 12$, $r = 0.85$. (a) Compute the least-squares regression line $\hat{y} = a + bx$. (b) Predict the improvement score for a patient who completes 8 weeks. (c) Compute r^2 and interpret it. (d) A residual plot shows a curved pattern. What does this suggest about the model?

2. A dataset of 50 cities records average summer temperature (x , °F) and annual ice cream sales per capita (y , dollars). The regression line is $\hat{y} = -120 + 4.5x$ with $r^2 = 0.61$. (a) Interpret the slope in context. (b) Interpret r^2 in context. (c) The warmest city in the dataset has $x = 95^\circ\text{F}$. Is using the line to predict sales at $x = 110^\circ\text{F}$ appropriate? Why or why not? (d) A journalist concludes: “Eating ice cream raises body temperature.” What is wrong with this reasoning?
3. Three scatterplots have correlations $r = -0.92$, $r = 0.45$, $r = 0.01$. (a) Match each r to its description: strong negative linear relationship; weak positive linear relationship; no linear relationship. (b) Compute r^2 for each. Interpret each as a percentage. (c) For $r = 0.01$: even though there is essentially no linear relationship, could x and y be strongly associated in a non-linear way? Explain.
4. Using the formula $I(X; Y) = -\frac{1}{2} \log_2(1 - r^2)$: (a) Compute I for $r = 0$, $r = 0.5$, $r = 0.9$, $r = 0.99$. (b) As $r \rightarrow 1$, what happens to I ? Interpret this. (c) Compare to the mutual information calculations in Chapter 9. What is the common idea?
5. *The final question.* Write a paragraph (4–6 sentences) that traces the single central question of this course — “what does one variable tell you about another?” — from Chapter 1 (standard deviation, a single variable described) through Chapter 10 (regression, r^2 , the connection to mutual information). Name at least five tools from the course and explain how each contributes to answering this question.